
Part V

Concurrent Aerobic and Strength Training for Athletic Performance



Long-Term Effects of Strength Training on Aerobic Capacity and Endurance Performance

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Introduction

Strength training is currently regarded as an important part of endurance athletes' long-term training program. However, the scientific evidence on long-term effects of concurrent strength and endurance training is sparse and only a few strength training interventions in endurance athletes last more than 15 weeks. This chapter: (1) summarizes the current scientific evidence on the effects of long-term strength training (i.e., defined as interventions lasting above 12 weeks) on aerobic capacity and endurance performance, and (2) discusses the potential neural and cellular mechanisms related to these effects, in order to provide a framework for generating new hypotheses in this exiting area.

Scientific Evidence

The requirements of strength for endurance performance in sport depend on the competitive duration and format, as well as the exercise mode employed. Endurance training is the essential stimulus for endurance athletes and they do traditionally not prioritize high volumes of strength training (i.e., training aimed to increase the muscles ability to generate maximum force). However, as highlighted in other chapters in this book, recent research has demonstrated positive short-term intervention effects of supplementing strength training concurrent to endurance training on aerobic capacity and performance. These studies have shown that aerobic capacity is mainly enhanced by improvements in exercise economy/efficiency and delayed fatigue when strength training is added. Partly because of the rising evidence

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showing its positive effects, strength training has become part of the training regime of many successful endurance athletes over the last years. However, since the majority of training interventions lasted only 8–12 weeks, caution should still be used when long-term effects are considered. We do currently not know the exact impact of long-term strength training on endurance performance. However, based on knowledge from power-based sports, it is logical to assume that progression and periodization in load, velocity, and/or changes in types of exercises would stimulate further long-term adaptations in strength in endurance athletes aiming to optimize their aerobic capacity.

The longest strength training intervention on endurance athletes to date is done by Rønnestad et al. [1], who performed a 25-week intervention in highly endurance trained cyclists. This study showed substantial positive effects on maximal strength and power variables, as well as power output at 4 mmol/L blood lactate concentration and maximal aerobic power. However, the effect sizes are in line with studies of shorter duration, and no mid-test was performed to analyze the effect of intervention duration on subsequent responses. A recent meta-analysis by Beattie et al. [2] confirms that, aside from Rønnestad et al.'s [1] 25-week strength intervention, the intervention periods used in the literature were 16 weeks or less, and the average duration approximately 10 weeks. Hence, much of what we know about neurological and structural adaptations from strength training derives from short-term interventions, and otherwise the interventions involve relatively untrained or inexperienced subjects. In fact, there are only a few studies investigating the long-term strength adaptations in well-trained athletes; however, these are from strength and power sports, e.g., rugby players [3]. Future research in well-trained endurance athletes should focus on long-term strength interventions and the subsequent changes in endurance performance.

Interestingly, Berryman et al.'s [4] recent meta-analysis of strength training on well-trained endurance athletes revealed a significant effect of the duration of the strength training, where protocols including more than 24 sessions led to greater improvements in work economy compared to protocols of less than 24 sessions. In addition, one case study used a scientific approach to combine strength and endurance training and, additionally, measured both strength and endurance variables systematically over the annual cycle [5]. In this study, the authors found an increase in 6RM leg press over 10 weeks before the progression plateaued over the rest of the year. Although there were further improvements in peak aerobic power over the year, it is unknown whether the strength training or other types of training affected this.

Another important question is whether the effects from an intensive period of strength training would be maintained in subsequent periods without or with less strength training. This is highly interesting with relation to periodization of strength training over the annual training cycle. Research shows that only a small part of the strength gained during an intervention is maintained after 8–12 weeks without strength training, which is accompanied by a relatively rapid reduction in muscle cross-sectional area and peak power output. Even after the 25-week training intervention by Rønnestad et al. [1], there was a rapid decline of adaptations during the

subsequent 8 weeks. Thus, in the long-term periodization of strength training for endurance athletes, “maintenance training,” as previously shown effective by Rønnestad et al. [6], seems to be required. However, strength training over longer periods often have to replace other types of training in highly trained athletes with already full training schedules to avoid overtraining. It can be argued that replacement of endurance training with strength training may have negative effects on endurance capacity. However, this was not the case in the study by Paavolainen et al. [7], who replaced endurance training by a mix of plyometric and explosive strength training and found beneficial effects over a relatively short intervention period (around 10 weeks). However, no studies have currently examined whether such a replacement would positively or negatively affect overall performance over longer time-scales. Thus, the question is not only what the possible positive effects of strength training is, but also whether this compromises with a negative effect due to less of the replaced training.

There are currently no reports of a negative effect of heavy strength training on exercise economy, maximal oxygen uptake or lactate threshold in endurance sports. Furthermore, a study performed on top-level endurance athletes did not observe a negative effect after 16 weeks of concurrent heavy strength training and endurance training on muscle capillarization [8], and after a period of concurrent strength and endurance training, there seems to be no impairment of the oxidative enzyme activity in endurance-trained athletes [9, 10]. Thus, with regard to muscle vascularization and oxidative potential, there seems to be no indications of negative effect of strength training—at least not when strength training is added over a shorter period. In fact, increased muscle mass without any reduced capillarization would in theory increase the overall oxidative potential of the muscle and thereby improve endurance performance. In this setting, muscle hypertrophy can be a “long-term” mechanism for improved aerobic capacity and Vikmoen et al. [11] recently reported a correlation between increased muscle mass and fractional utilization of VO_2max (and performance) in female cyclists.

Even though strength training can be added to endurance training without a concomitant increase in total body mass, there seems to be a small, ~3–6%, increase in measurements of muscle hypertrophy in the main target muscles over 10–12 weeks [6, 8, 12, 13]. If this trend continues over longer strength training regimes, it might potentially have some negative side-effects, especially if strength exercises increase the mass of non-functional muscles. It can therefore be assumed that it is particularly important that strength-training exercises should involve similar muscle groups and imitate the sports-specific movements when performed over longer time-periods.

Based on the overall scientific evidence, endurance athletes are currently advised to build up maximal strength in the important muscles during the preparatory period by two strength training sessions per week, using a “daily undulating periodized program” [14]. Loads between 4RM and 10RM are typically used, with 2–3 sets per exercise and approximately 2–3 min of rest between sets. However, during the subsequent training periods and even during the competitive season, where development of strength is not prioritized, approximately one strength training session per

week (low volume) with high intensity seems to be needed to maintain the previous strength training adaptations [6]. However, for all these aspects, the athletes' training history, modality of aerobic training, and intervention duration might represent important variables potentially explaining individual effects. Since these recommendations are mainly based on short-term studies combined with practical work with elite endurance athletes, future research should be conducted to study the actual effects of different long-term periodization strategies. This would be helpful to provide the practitioner with more precise guidelines regarding, for example, the appropriate timing for the implementation of strength development within the annual training plan for different individuals.

Although sport-specific requirements of strength and endurance are relatively well understood in many endurance sports, we understand less of the optimal mix of strength and endurance training puzzled into the daily training plan of elite endurance athletes. How should it optimally be periodized in micro and macro cycles with changing volume of strength versus endurance training? How could employment of different muscle groups in strength and endurance training be done? Should different types of nutrition and overall energy availability be prioritized before, during, and after strength vs endurance training? These are examples of areas that are relatively unexplored, at least in a long-term perspective. The reality might be that science has not caught up with the sophistication of modern training and recovery techniques used by the best athletes. For example, world-class rowers or XC skiers can build up and sustain a large and strong muscle mass while, at the same time, developing an outstanding aerobic endurance capacity.

Potential Neural and Cellular Mechanisms

Although there is currently limited scientific evidence for the positive effects of long-term strength training on endurance performance, many of today's top athletes in endurance sports use various types of strength training over longer time-scales in order to enhance performance. Together with the results from short-term studies, we therefore suggest that similar effects are present also over longer time-spans although progression and periodization of both strength and endurance training is possibly needed to gain long-term progress. However, as with other types of training, it is likely that the acceleration of these effects reduces over time, as the body adapts to the "new" load. In the following, some of the main long-term mechanisms that may be present when endurance athletes supplement their endurance training with heavy strength training will be discussed. Still, the reader should be aware of the limited evidence in this area and researchers should use this to generate new hypotheses that can be tested in future studies.

The muscle fiber recruitment pattern may change with long-term strength training. Strengthening the type I fibers may lead to lower relative activity of the muscle and postpone their fatigue over long-duration work, thereby also postponing the employment of type II fibers that are normally less economical. The type II fibers can subsequently be used later, which may be the reason why some studies show

positive effects of strength training only after 2 h of submaximal cycling or skiing [15]. A reduced reliance of type II fibers may also increase economy through longer maintenance of glycogen stores that subsequently increase performance after prolonged exercise. However, long-term strength training may also increase the portion of type IIA fibers, thereby reducing type IIX, as previously shown by Aagaard et al. [8]. The type IIA fibers are more fatigue resistant, but still have the possibility to produce relatively high power. The findings in this area are conflicting [9], so future research needs to confirm the potential effect.

Increased maximum force and/or increased rate of force development capability following strength training may facilitate better blood flow to the working muscles [8, 14]. Increased rate of force development is likely caused by increased neural activation and has been linked to improvements in movements/sports where faster force development is positive for performance, e.g., to improve force effectiveness, leading to less relative load on the muscle or longer recovery phases during a cycle, thereby inducing less time with constriction of the blood flow. Although this has not yet been shown directly, we suggest that it would lead to improved delivery of oxygen and substrates during exercise.

Another area with conflicting theories is how strength training may influence the activation of muscle fibers during endurance exercise. One theory says that less muscle mass can generate the same work when the muscle is stronger, thereby reducing the oxygen cost of exercise. Other theories say that the same muscle mass can do the same work at lower relative activation levels when stronger, subsequently improving the ability to increase power at the end of an endurance competition. However, long-term adaptations might occur also on the muscle-tendon level. In some sports, such as running, increased muscle-tendon stiffness due to strength training may improve performance for some individuals [14]. However, this effect is influenced by the technical demands of running and the baseline stiffness of the individual. In other sports, such as cycling, this mechanism is unlikely to have an effect on performance.

There are also cellular effects of strength training occurring on a long-term perspective, such as activation of the phosphatidylinositol 3-kinase (PI3-k)-Akt-mammalian target of rapamycin (mTOR) signaling pathway. This is regarded to regulate the rate of protein synthesis and may, with long-term training, induce muscle hypertrophy. However, activation of adenosine monophosphate-activated protein kinase (AMPK) caused by high levels of endurance exercise may inhibit mTOR and suppress myofibrillar protein synthesis. Thus, large volumes of endurance exercise may negatively affect intracellular pathways of importance for myofibrillar protein synthesis [16], and placing strength sessions optimally will be especially important for endurance athletes in periods where the goal is to increase muscle mass. This also highlights the importance of sufficient amount of strength training to maintain muscle mass and strength in periods with high volumes of endurance training or competitions. However, note that the molecular research in the area is in its infancy and can currently not be directly applied to the physical preparation of endurance athletes.

While sufficient mechanical loading is important for gaining optimal strength training effects, also the movement pattern during training is important. This should

be similar to the movement in the corresponding discipline, to trigger specifically the optimal activation patterns of the involved muscles and inducing cellular adaptations. In addition, training models should aim to avoid strength- and endurance-training interference during periods of concurrent aerobic and strength training by planning training with awareness of the possible molecular interference and by including optimal nutrition. Overall, these different mechanisms should likely be triggered in different individuals and different sports, depending on the demands of the sport in relationship to the individual's capacity. This subsequently influences the choice of load, series, repetitions, and type of exercise that should be chosen for a given athlete—indicating that there is no “ideal” strength training program for a give sport but rather a toolbox that coaches and athletes should choose from in different contexts.

Summary

Overall, both scientific and practical evidence suggest that strength training could be successfully included in elite endurance athletes' training program, twice weekly to build up strength in the preparation period and once per week to maintain strength in the competition period. By using sport-specific strength programs, aerobic capacity seems to mainly be improved through exercise economy/efficiency and sometimes by delayed fatigue. Currently no negative effects of strength training on aerobic capacity has been shown. The possible neural and cellular mechanisms include: (1) a change in the muscle fiber recruitment pattern that postpones the muscle's fatigue over long-duration work and leads to longer maintenance of glycogen stores, (2) possible increase in the portion of type IIA fibers that are able to combine high strength and endurance capacities simultaneously, (3) increased rate of force development that facilitates better blood flow to the working muscles, thereby improving delivery of oxygen and substrates during exercise, (4) increased muscle-tendon stiffness in movements where this is beneficial, and (5) maintained signaling from intracellular pathways of importance for myofibrillar protein synthesis, induced by a training pattern with low negative inference from the large amounts of endurance training. However, the effects of long-term strength training on these factors, as well as how progression and periodization strategies can be used to prolong the positive effects of strength training on endurance performance, are not shown in scientific studies and need to be further examined.

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